Anatole Lefort

✉ anatole.lefort@gmail.com
anatolelefort.net
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Research Interests

Distributed Systems, Operating Systems, Concurrency, Persistent Memory, Language Runtimes, Cloud Infrastructures, Cache Coherence, Coherent Fabrics (CXL)

Education

2018 - 2023 **Ph.D. in *Computer Science***, *Institut Polytechnique de Paris*, Palaiseau (91), France.
Thesis: *"A Support for Persistent Memory in Java"* – Advisors: *Pierre Sutra, Gaël Thomas.*

2015 - 2018 **Dipl. Ing. (Msc), in *Computer Science***, *Télécom SudParis*, Évry (91), France.
Majors: *Distributed systems and computing* – Graduated first in class.

2013 - 2015 **MPSI/MP* (CPGE)**, *Lycée Camille Guérin (86)*, France.
Intense & highly « competitve exam preparatory courses », for admission to french higher-education national engineering schools. Majors: *Mathematics, Physics*; Minor: *Computer Science*

2010 - 2013 **Science Baccalauréat**, High-School: *Lycée Pilote Innovant et International (86)*, France.
With honours, European class in engineering sciences, Majors: *Mathematics, Engineering sciences*

Experience

Research

Oct 2023 - *Present* **Humboldt Post-doctoral Researcher, *Technical University of Munich***, Germany.
Group: *Chair of Computer Systems* – Host: *Pramod Bhatotia*
- **CXL memory consistency:** shared remote CXL memory across heterogeneous architectures.
- **CXL bridges:** integrating CXL 3.0 cache coherence into arbitrary host architectures.
- **CXL GPUs:** GPUs with CXL 3.0 integration for symmetric cache coherence with host CPUs.

Oct 2018 - Feb 2023 **Graduate Research Assistant, *Institut Polytechnique de Paris***, France.
Group: *Parallel and Distributed Systems* – Advisors: *Pierre Sutra, Gaël Thomas*
- **J-NVM:** a pure-java library to efficiently support persistent memory in Java.
- **NVMM & FaaS:** persistent memory for stateful serverless workloads.

Mar 2018 - Aug 2018 **Research Intern, *IMDEA Software***, *Madrid*, Spain.
Advisor: *Alexey Gotsman*
- **White-box atomic multicast:** a novel & efficient distributed protocol for genuine atomic multicast. Implemented in *C* with *libevent*, evaluatuated in *local* and *geo-wide* area networks.

Undergraduate Research Projects

Fall 2017 **Scalevisor**, *a research microkernel and bare-metal hypervisor for NUMA machines.*
Master 2 project at *Télécom SudParis*, 2-student groupwork – Advisor: *Gaël Thomas.*
- Port of the microkernel from x86 AMD platform to Intel's. (*C, x86 assembly*)
- Add support for vendor-specific virt. extensions (*Intel VMX*), replacing *AMD-V* & *AMD-Vi.*

Spring 2017 **Mesosearch**, *investigate dynamic service scaling for orchestrated containers.*
Master 1 project at *Télécom SudParis*, 4-student groupwork – Advisors: *Pierre Sutra, Gaël Thomas.*
- Deployment of a real-world distributed test app – a fully scalable and elastic web search engine using *Apache Mesos, Zookeeper, HDFS, Hadoop, Myriad, Nutch and Elasticsearch.*
- Implementation of an application-oriented resource monitoring system for Mesos, providing necessary data to enable automatic and dynamic scaling.

Student Volunteering

Oct 2015 - Jun 2019 **Sys & Net Admin at MiNET**, *Student Internet Service Provider*, Évry (91), France.
Volunteer in the free-software student association. Internet access provider and network infrastructure for the on-campus housing. Approx. 850 rooms in 7 buildings, within a team of 15 admins.
- **IT support:** help-desk activities and free-software promotion to all students on campus.
- **Event organization:** Linux install party, free software & technical annual conference, massive triennial student night party with more than 1000 attendees.
- **IT Admin:** *(SYS)* Debian, Proxmox VE, *(NET)* Cisco, *(Software Dev)* Ruby, Python, Bash.

May 2016 - May 2017 **Training supervisor at MiNET**, *Student Internet Service Provider*, Évry (91), France.
In charge of technical trainings for new members (student administrators).

Publications

Conference

MICRO'26 Nominated for Distinguished paper award
Synapse: Decoupling Protocol, Topology, and Policy for Automated Synthesis of Coherence Controllers
Nicolò Carpentieri, **Anatole Lefort**, David Schall, Julian Pritzi, Pramod Bhatotia.
In *59th IEEE/ACM International Symposium on Microarchitecture* (MICRO), Athens, Greece, 2026

ASPLOS'26 Best paper award
vCXLGen: Automated Synthesis and Verification of CXL Bridges for Heterogeneous Architectures
Anatole Lefort, Julian Pritzi, Nicolò Carpentieri, David Schall, Simon Dittrich, Soham Chakraborty, Nicolai Oswald, Pramod Bhatotia.
In *Conference on Architectural Support for Programming Languages and Operating Systems* (ASPLOS), Pittsburgh (PA), USA, 2026

HPCA'26
C^3: CXL Coherence Controllers for Heterogeneous Architectures
Anatole Lefort, David Schall, Nicolò Carpentieri, Julian Pritzi, Nicolai Oswald, Pramod Bhatotia.
In *Symposium on High-Performance Computer Architecture* (HPCA), Sydney, Australia, 2026

SOSP'21
J-NVM: Off-heap Persistent Objects in Java
Anatole Lefort, Yohan Pipereau, Kwabena Amponsem, Pierre Sutra, Gaël Thomas.
In *Symposium on Operating Systems Principles* (SOSP), Virtual, 2021

DSN'19
White-Box Atomic Multicast
Alexey Gotsman, **Anatole Lefort**, Gregory V. Chockler.
In *Dependable Systems and Networks* (DSN), Portland (OR), USA, 2019

Journals

TOCS
CXL Bridges for Multi-Host Coherent Memory across Heterogeneous Architectures
Invited extended journal version through recommendation by the PC chairs of ASPLOS '26
Anatole Lefort, Julian Pritzi, Nicolò Carpentieri, David Schall, Simon Dittrich, Soham Chakraborty, Nicolai Oswald, Pramod Bhatotia.
In *ACM Transactions on Computer Systems* (TOCS), forthcoming, 2026.

Talks

Conference Talk In *Conference on Architectural Support for Programming Languages and Operating Systems* (ASPLOS), Pittsburgh (PA), USA, 2026, *vCXLGen: Automated Synthesis and Verification of CXL Bridges for Heterogeneous Architectures*

Conference Talk In *Symposium on High-Performance Computer Architecture* (HPCA), Sydney, Australia, 2026
C^3: CXL Coherence Controllers for Heterogeneous Architectures

Conference Talk In *Symposium on Operating Systems Principles* (SOSP), Virtual, 2021
J-NVM: Off-heap Persistent Objects in Java

Workshop Paper In *Non-Volatile Memories Workshop* (NVMW), San Diego (CA), USA, May 2022
J-NVM: Off-heap Persistent Objects in Java.

Invited Talk In *Heterogeneous Memory Workshop* (HMEM), Virtual, July 2022
J-NVM: Off-heap Persistent Objects in Java

Professional Service

- Program Committee member **EuroSys'27** main research track, *spring and fall* deadlines
- Program Committee member **SOSP'26** posters track, and *on-site poster session organization*
- Program Committee member **HCDS'26** (*Workshop on Heterogeneous Composable and Disaggregated Systems*)
- Artifact Evaluation Committee of **SOSP'23**

Prizes and Awards

- **ASPLOS'26 Best paper award**
  for the paper *vCXLGen: Automated Synthesis and Verification of CXL Bridges for Heterogeneous Architectures.*
- **Laureate of "Engineers of the Future Awards"**, *Research Engineer category, 2022 Edition.*
  Sponsored & Issued by *L'Usine Nouvelle* - a renown french business magazine.
- **Best student publication in ICTs at Institut Polytechnique de Paris**, *2022 Edition.*
  Sponsored & Issued by *Labex DigiCosme*, *Institut Polytechnique de Paris* and *Université Paris-Saclay.*

Grants

- **Humboldt Research Fellowship for Postdocs** from *the Alexander von Humboldt Foundation*, 2023-2025.
- **NVMW Student Travel Grant** to attend *NVMW'22* (San Diego, CA, USA).
- **Fully funded Ph.D. scholarship** from *Institut Mines-Télécom*, *"Future & Rupture"* campaign (2018). Awarded on *Academic Excellence* criteria, ranked 1st for *Télécom SudParis*.

Teaching Experience

- **Practical Lab: System Programming (C, Rust)**:
  Co-coordinator Winter 2024-25, Summer 2025
  Teaching Assistant Winter 2023-24, Winter 2025-26, Winter 2026-27
- **Lecture: Introduction to Software Engineering (Java)**:
  Lecturer & Exam writing Summer 2024, Summer 2026
- **CSC 3102: Operating systems, Unix & Shell scripting (Bash)**:
  Lecturer Fall 2020
  Teaching Assistant Fall 2019
- **CSC 3101: Data structures & algorithms (Java 1)**:
  Teaching Assistant Fall 2020
- **CSC 4102: Object-oriented programming & software engineering (Java 2)**:
  Teaching Assistant Spring 2020, Spring 2021

Student Supervision

- Nicolò Carpentieri, PhD, Fall 2024 - ongoing, *Coherence Controllers for Heterogeneous Architectures*
- Nguyen Le, Bsc, Summer 2026, *Heterogeneous Litmus Tests for CPU-GPU Memory Consistency*
- Ivan Logvynenko, Bsc, Summer 2026, *Flat Memory for CXL 3.0 GPUs*
- Victor Trost, Bsc, Winter 2025, *Disaggregated and Vendor-Generic APUs with CXL*
- Julian Pritzi, Msc, Winter 2024, *Formal Verification of Heterogeneous Cache Coherence Protocols for CXL*
- Simon Dittrich, Msc, Winter 2024, *Generating Heterogeneous Cache Coherence Protocols for gem5 Simulations*
- Mouhamadou Mangassouba, Msc, Spring 2022, *P-FaaS: Stateful Serverless with Persistent Memory*
- Kwabena Amponsem, Msc, Fall 2020, *P-Transformer: A Java Bytecode Transformer for Persistent Objects*

Skills

Computing

Programming Java, C, Python, Bash, SQL, Latex, Go, OCaml, B-method, PHP, Ruby

Systems Linux, Unix, Cisco IOS

IT Admin

Virt. Proxmox VE, QEMU/KVM, LXC, Docker

Services FreeRadius, Apache, Nginx, OpenVPN, Zabbix, ISC-KEA, ELK

Languages

English European C1 level - TOEIC 970

French Native speaker

German European A2 level

Spanish Beginner

Other Interests

Hobbies Computer tinkering (hardware and software); problem solving.
Freeline skating, Outdoor activities.

Sports Cycling (Road, MTB, Touring, Bikepacking),
Running (Road or Trail), Swimming
Hiking & Backpacking, Skiing, Snowboarding

Past Sports Gymnastics, Tennis, Judo